

- 4 (a) electrical potential energy (stored) when charge moved and gravitational potential energy (stored) when mass moved
 due to work done in electric field and work done in gravitational field B1
 B1 [2]
- (b) work done = force $\times$ distance moved (in direction of force)
 and force = mg M1
 $mg \times h$ or $mg \times \Delta h$ A1 [2]
- (c) (i) $0.1 \times mgh = \frac{1}{2} mv^2$ B1
 $0.1 \times m \times 9.81 \times 120 = 0.5 \times m \times v^2$ B1
 $v = 15.3 \text{ m s}^{-1}$ A0 [2]
- (ii) $P = 0.5 m v^2 / t$ C1
 $m / t = 110 \times 10^3 / [0.25 \times 0.5 \times (15.3)^2]$ C1
 $= 3740 \text{ kg s}^{-1}$ A1 [3]
- 5 (a) ohm = volt / ampere B1 [1]